10 DETERMINE THE VOIDS RATIO FOR THE GIVEN SAND SAMPLE

Aim

To determine the voids ratio for the given sand sample.

TJH3ZH

Definition

It is defined as the volume of voids to the volume of solids present in the soil. It has no units.

$$\text{Voids ratio 'e'} = \frac{\text{volume of voids}}{\text{volume of solids}} = \frac{V_v}{V_s} = \frac{V_g + V_w}{V_s}$$

Apparatus and Materials Required

1. Weighing Balance
2. Bucket
3. Sand
4. Water
5. 16mm dia, 600 mm length tamping rod

Procedure

1. Take a clean dry bucket and weigh it (W_1 grams).
2. Take about 1/3 rd bucket of clean sand and tamp it 25 times. Do the same twice to fill the bucket.
3. Weigh the bucket with sand (W_2 grams).
4. Now fill the bucket with sand, with water and weigh it (W_3 grams).
5. Empty the bucket, fill it with water and weigh it (W_4 grams).

Observation

1. Weight of the empty bucket (W_1 grams) =
2. Weight of the bucket + sand (W_2 grams) =
3. Weight of the bucket + sand + water (W_3 grams) =
4. Weight of the bucket + water (W_4 grams) =

$$\text{Voids ratio 'e'} = \frac{W_3 - W_2}{(W_4 - W_1) - (W_3 - W_2)} =$$

Where

V_v = Volume of voids
V_s = Volume of solids
V_g = Volume of air
V_w = Volume of water

Result

Voids ratio of the given sand sample is = _____

XI_Civil_Practical_English.indd 241 12/21/2019 5:50:42 PM